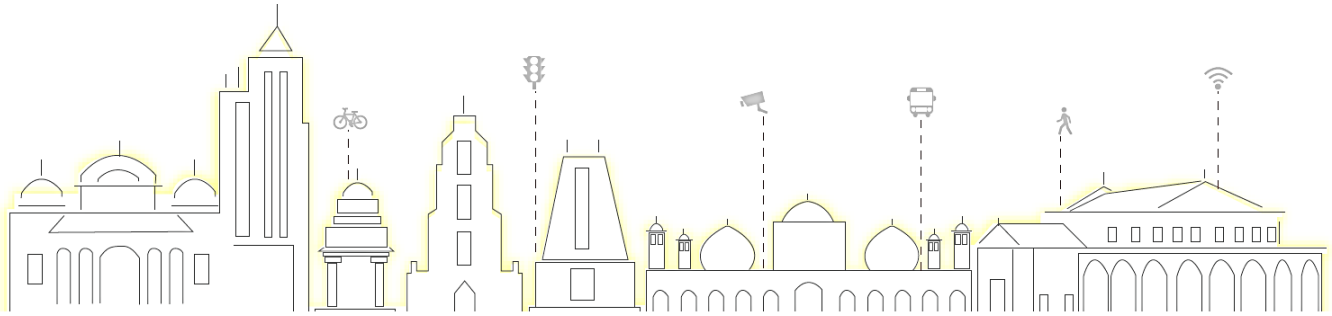
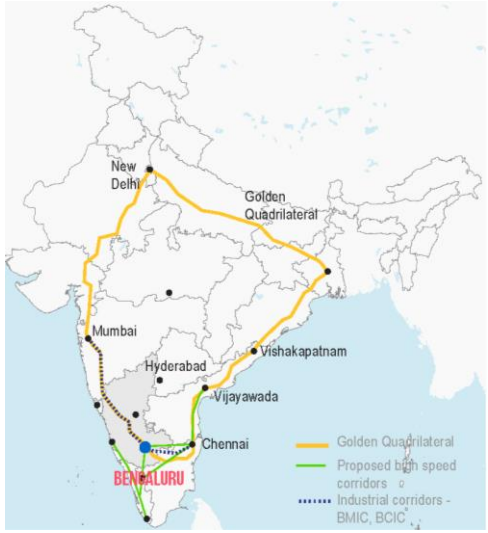


SMART CITY PROPOSAL, Bengaluru

CITY PROFILE

City context & Demographics



POPULATION
9.8 Million (2015)

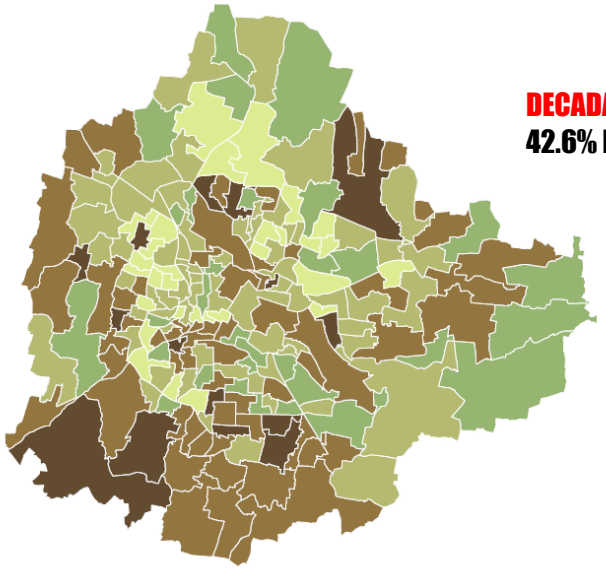
AREA OF CITY – 709.5 sq km
198 wards

POPULATION DENSITY
4378 persons/ sq km

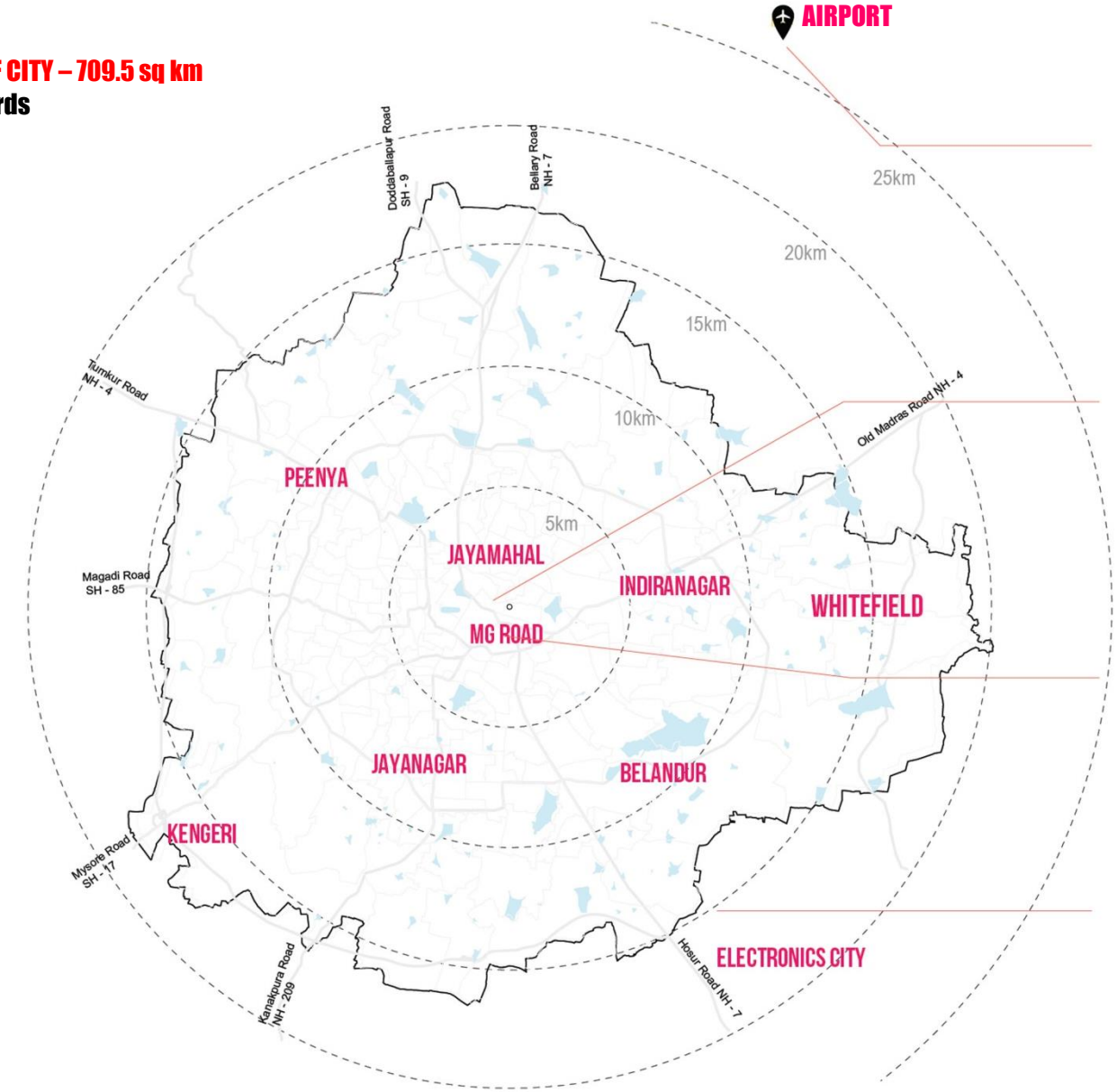
Literate males
3,595,059

Literate females
3,082,135

Sex ratio – 923 females for every 1000 males



DECADAL GROWTH RATE
42.6% Pan City



Kempegowda International Airport

Vidhana Soudha - State legislature building

Commercial Street and Brigade road

Electronics city - IT industry

CITY PROFILE

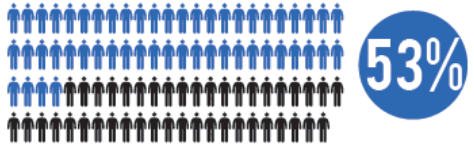
Transportation & Connectivity

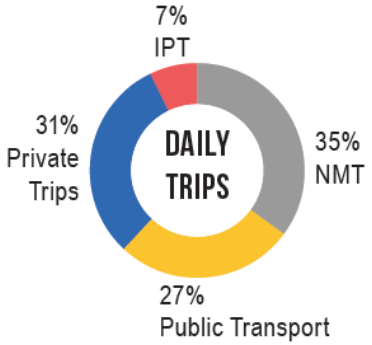
METRO NETWORK

42.3 km operational, 72.09 km proposed

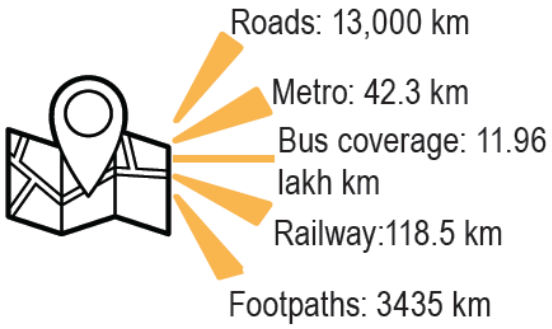
BUS PASSENGERS

5.02 million passengers





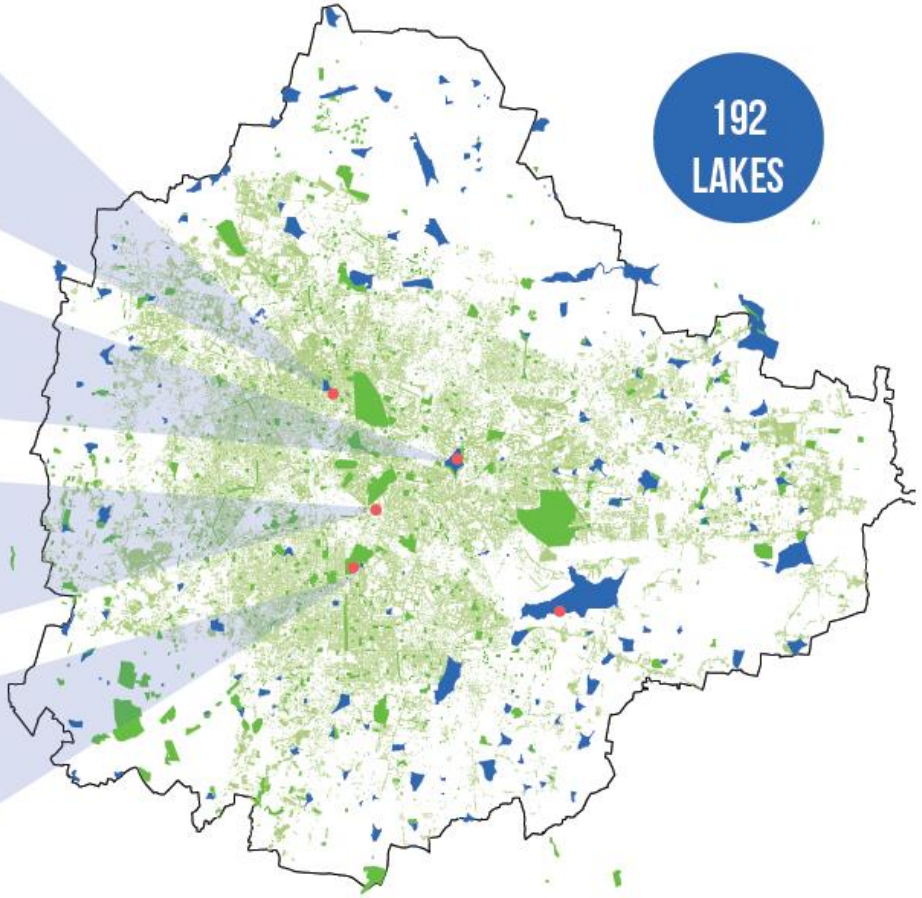
Source: mobility indicators, DULT -2010



Environment

LAKES & GREEN SPACES

17.5 sq km & 27.9 sq km

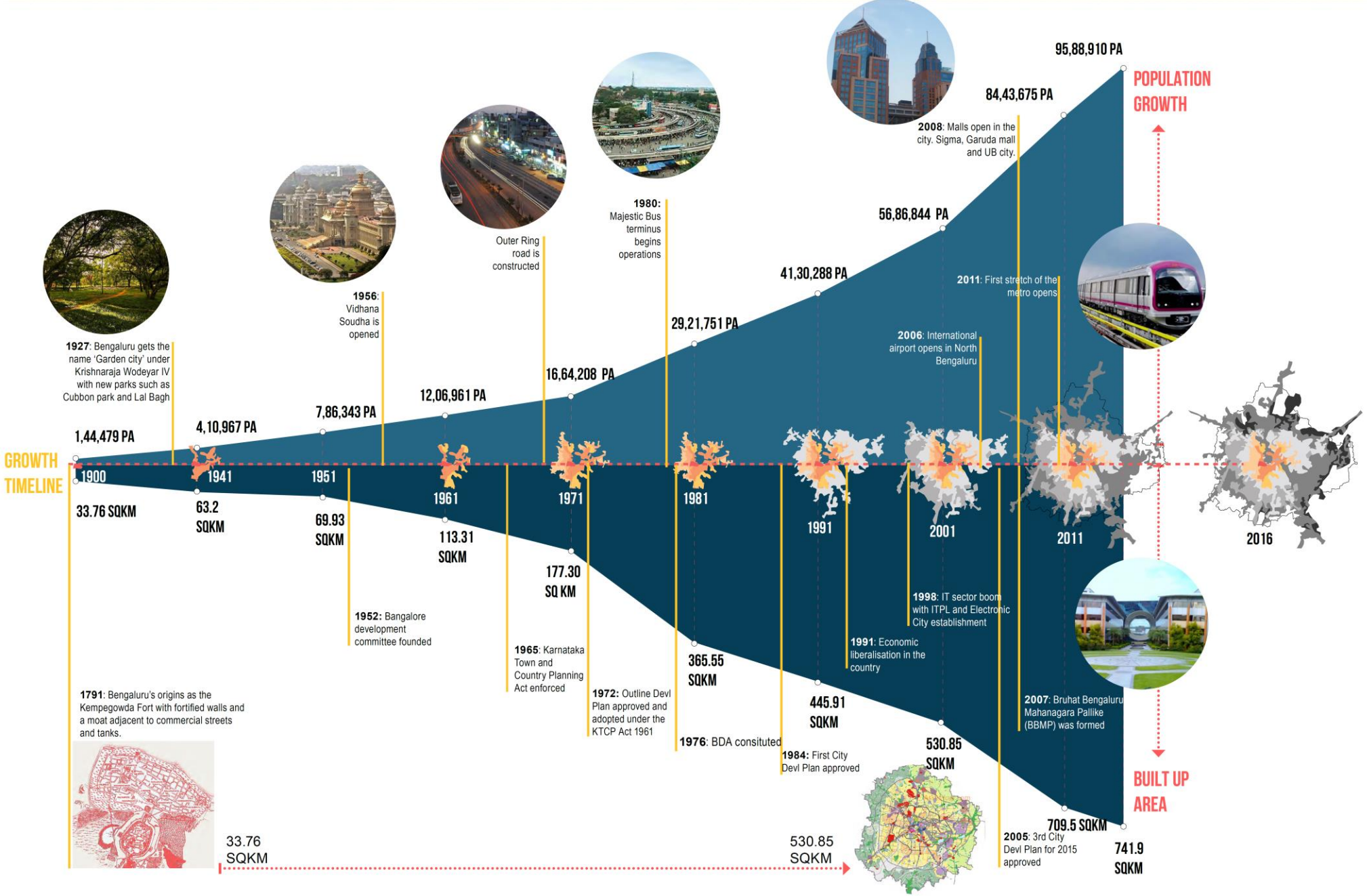


AREAS OF CONCERN

- **Severe traffic congestion**
- **Inadequate public transport facilities**
- **Inadequate pedestrian and disable friendly infrastructure**
- **Reduction of open spaces/ lakes**
- **Contamination of water bodies**
- **Sub optimal solid waste management**

CITY PROFILE

Evolution &



CITY PROFILE

Self Assessment

MOBILITY AND TRANSPORT

13000 km road network

2428 kms (18%) of public bus transit system

52L (53%) Daily ridership in BMTC

42.30 kms of Metro rail network

1% signalised intersection

26% footpath available

WATER SUPPLY

94 LPCD Water supply

38% coverage of water supply connection

50% quality of water supply

49% Non revenue water

95.5% metered connection
1400 smart meters

2% Houses with rain water harvesting facility

WASTE WATER & SWM

66.3% Coverage of Sewerage network

62.5% sewage treated in STPs

30 - 40% Segregation at source

60% collection at source

STORM WATER DRAINAGE

850 km Storm water drain

47% natural stream missing

38% Underground drainage network

SAFETY AND SECURITY

50% reduction in road accidents

21758 repeated offenders driving licenses sent for suspension

400 surveillance cameras

Source: Bangalore Traffic Police | Centre for Sustainable Development | BBMP | Street Quality Score

CITIZEN PARTICIPATION

15,20,414 RESPONSES

52 LAKH+ CITIZENS CONTACTED

TOTAL CAMPAIGN REACH

10.37 LAKH+ CITIZENS VOTED

CITY LEVEL VOTES

3 LAKH+ CITIZEN INPUTS

NEIGHBOURHOOD INPUTS

68,494 CITIZEN OPINIONS

SPECIFIC OPINIONS

ಬೃಹತ್ ಬೆಂಗಳೂರು ಮಹಾನಗರ ಪಾಲಿಕೆ
Bruhat Bengaluru Mahanagara Palike

Smart City
Namma
Bengaluru
Vote Now!

LIKES ON FB
6,24,891

TWEETS 5681

WHATSAPP MSGS
25,435

49,229 LIKES ON
SOCIAL MEDIA

2,52,422 DIRECT
BALLOTS

1,12,433 MISSED
CALL VOTES

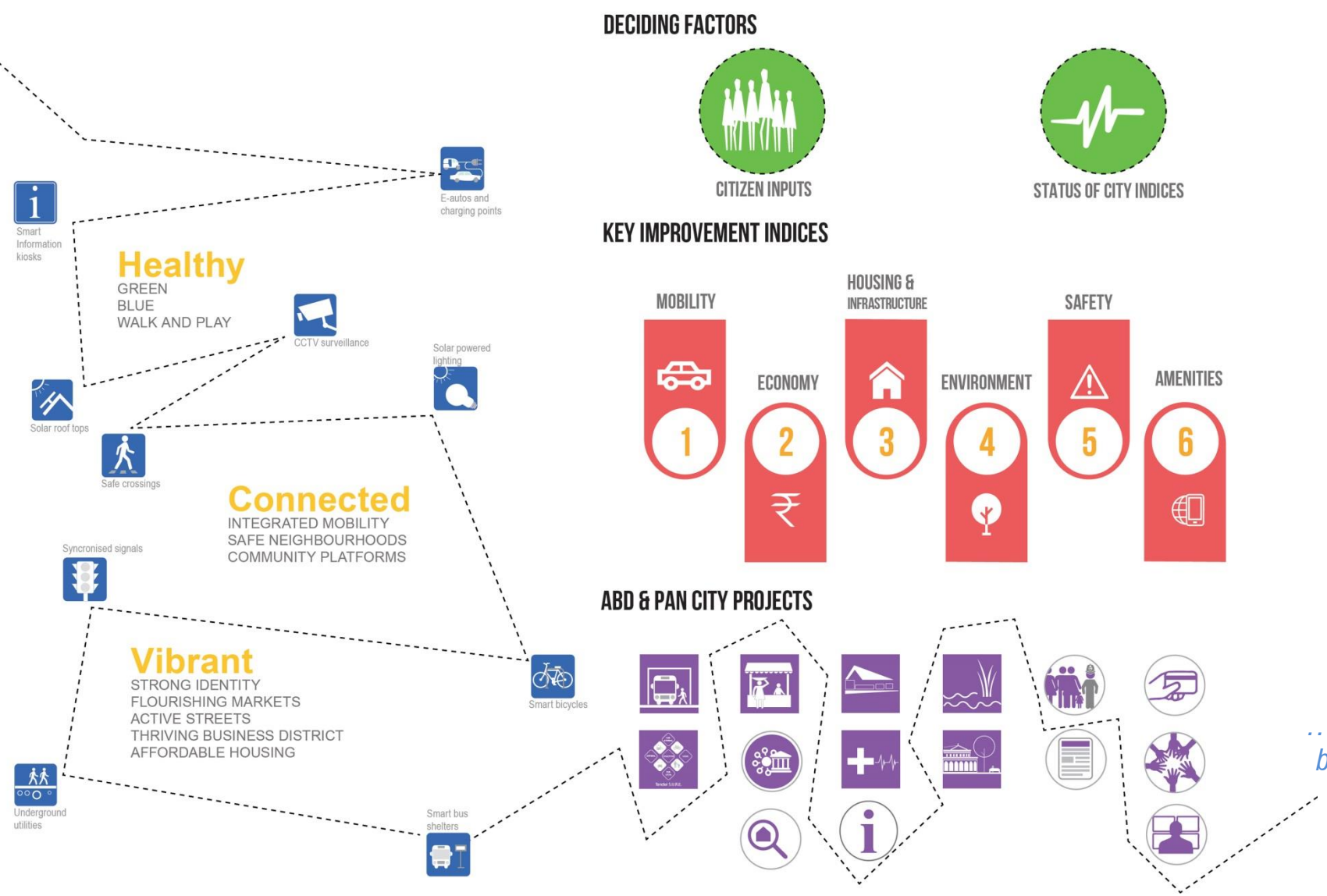
50 LAKH SMS SENT

YOUTUBE VIDEOS

67,114 INPUTS ON THE
BUDGET PRIORITIES

291 INPUTS ON
MYGOV.IN

Livable Bengaluru



…And, a smart city is built by building smart communities

PROPOSED INTERVENTIONS

ABD & Pan City

7 AREA BASED DEVELOPMENT PROJECTS

10 areas of interventions

PROJECT 1: **Revitalization of historic heart of city**
- Road redevelopment under Tender S.U.R.E. model

PROJECT 2. **Integrated mobility towards creating vibrant destination**
- Shivajinagar bus station and Russell Market
- Kempegowda Bus Stand (Majestic)

PROJECT 3: **Redevelopment of historic economic centres**
- KR Market
- Malleshwaram Market

PROJECT 4: **Innovation of downstream clean up of drainage system**
- Ulsoor lake
- Sankey Tank

PROJECT 5: **Protection & redevelopment of centrally located parkland**
- Cubbon Park

PROJECT 6: **Increasing affordable housing stock through slum redevelopment**
- Swathanthra Palya Slum redevelopment, Gandhinagar

PROJECT 7: **Retrofitting of a health care facility**
- K.C General Hospital

1 PAN CITY PROJECT

8 proposals on governance and public services

PROJECT 1: **CITIZEN CENTRIC E-GOVERNANCE AND SERVICES**

GOVERNANCE
- Municipal finance reforms
- Participatory budgeting
- Improvement in property tax collections
- Grievance management
- Online project information system for public project management
- Improvement of neighborhood safety

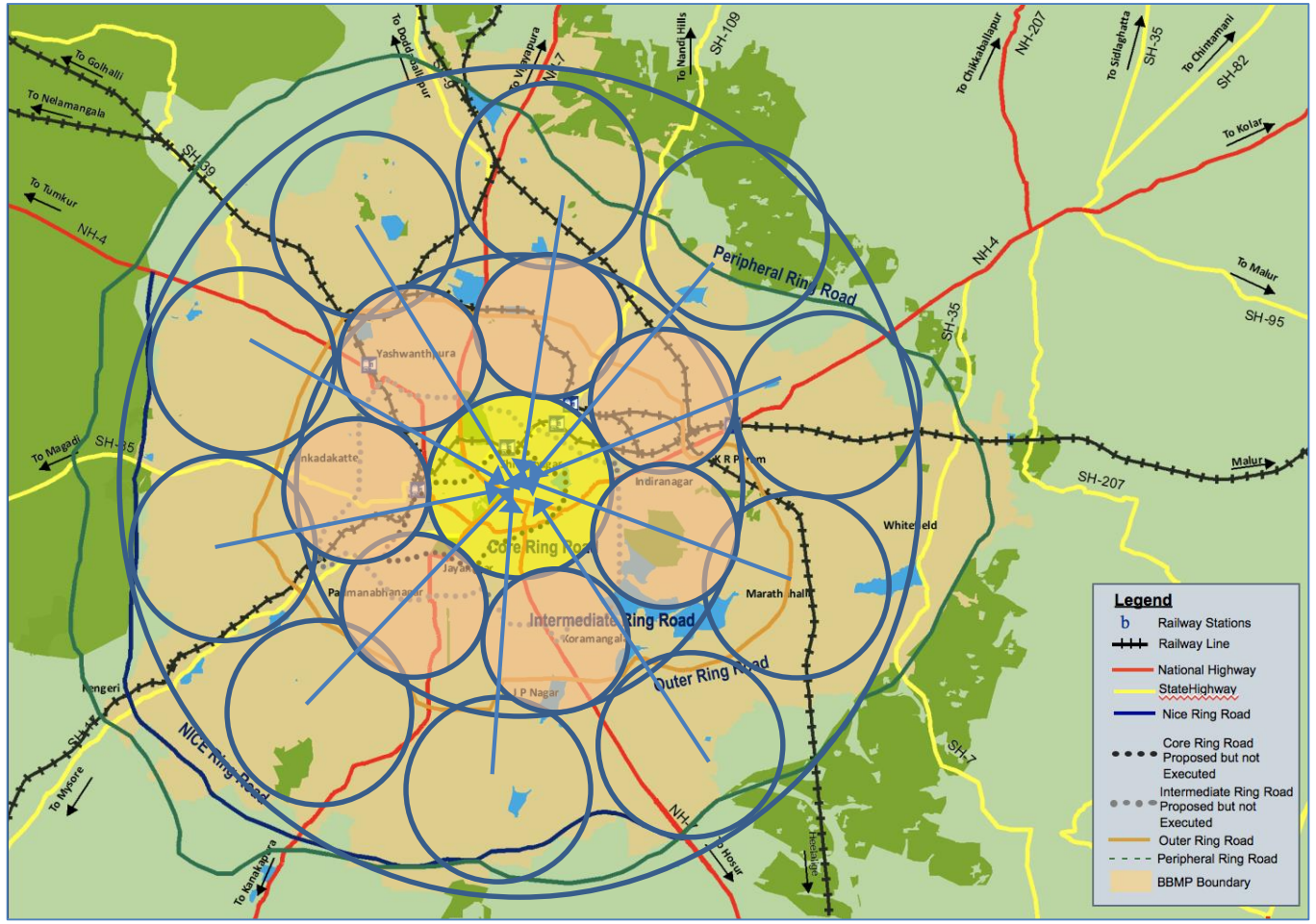
PUBLIC SERVICES
- BTRIPS – Bengaluru Travel Related Information and Planning System
- Open data portal for citizen information and innovation

AREA BASED DEVELOPMENT

Revitalizing the city center of Bengaluru

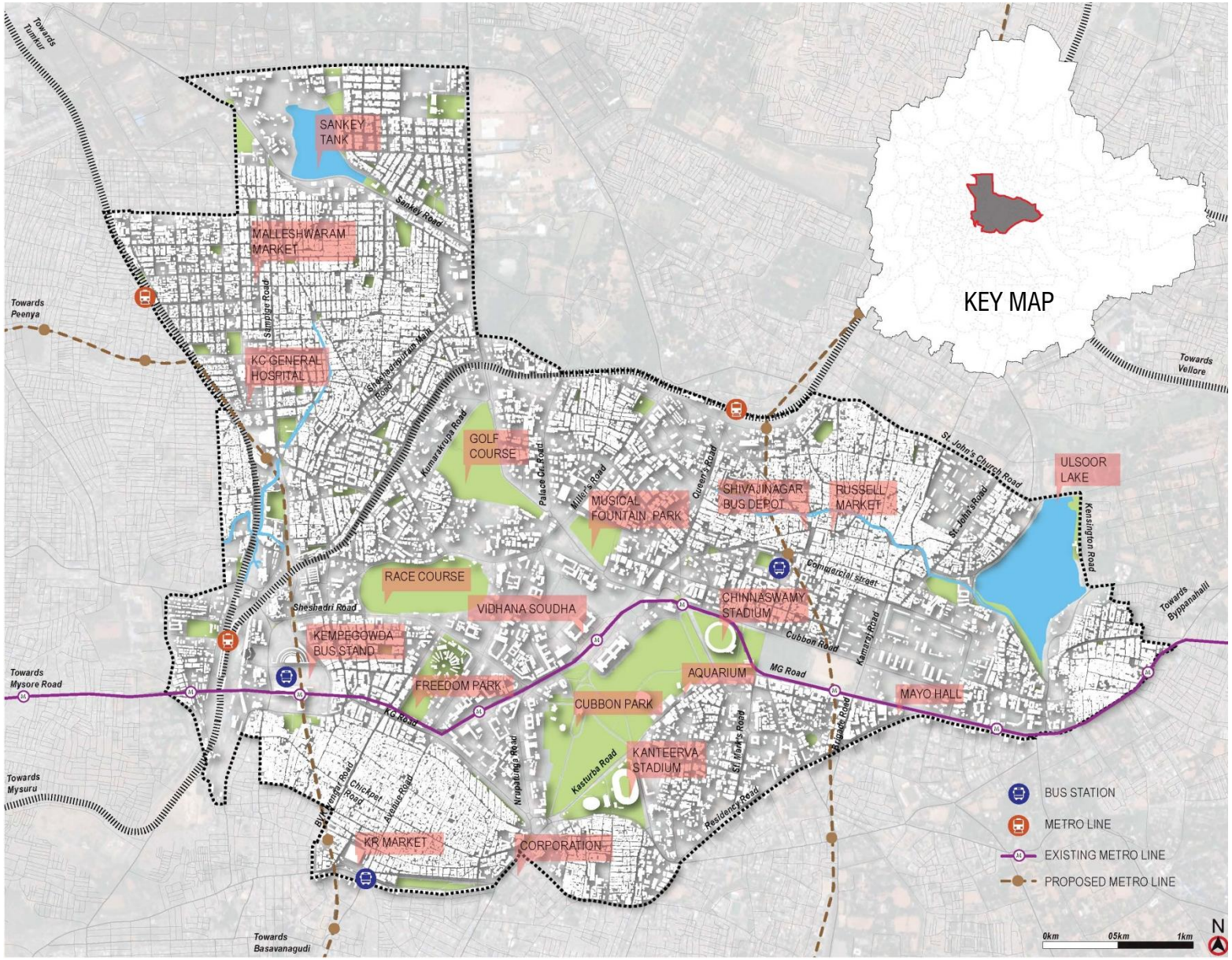
Area for maximum impact

"BELONGS TO, AND IS USED BY EVERYONE"



AREA BASED DEVELOPMENT

Revitalizing the city center of Bengaluru



AREA: 5380 ACRES (21.8 SQ KM)
POPULATION : 5,24,000
17 WARDS

ABD LANDMARKS

CBD & ADMIN
- Vidhana Soudha
- City corporation
- MG road
- Kasturba Road

OLD CITY CORE
- Historic Pete - Chickpet
- Avenue Road
- Cantonment
- Brigade road
- Commercial street

SOCIO-CULTURAL
- Kempegowda Museum
- Vishveshwariah Tech. Museum
- JN Planetarium
- Aquarium
- Chinnaswamy stadium

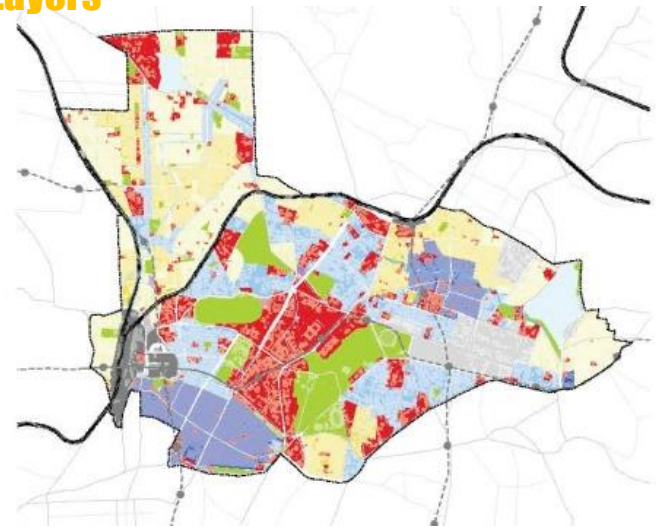
PARKS AND LAKES
- Cubbon Park
- Bal Bhavan
- IG Musical fountain
- JN Planetorium
- Ulsoor Lake
- Sankey Tank

CULTURE & HERITAGE
- Bangalore Fort
- Historic markets
- St.Mary's Basilica
- Jama Masjid
- Temples in Malleshwaram & Ulsoor

TRANSIT HUBS
- 2 city bus stations
- 2 metro interchanges and city railway station

AREA BASED DEVELOPMENT

City Layers



LAND USE MAP

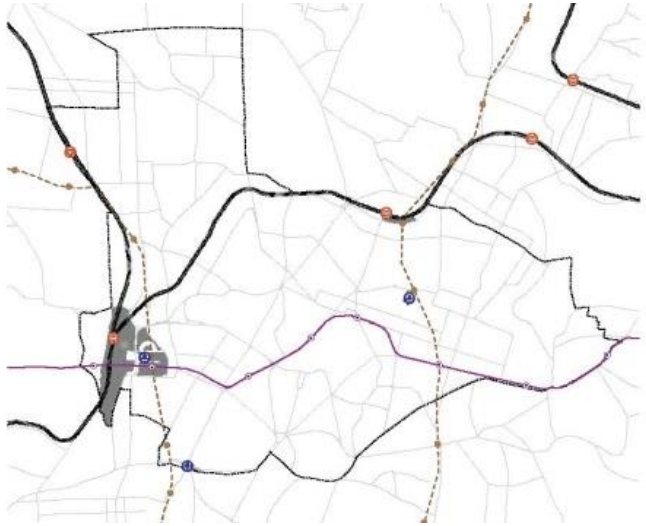
28% Commercial, 19% Public & Semi Public , 13% Mixed Use

BUILT vs OPEN

29% Built up

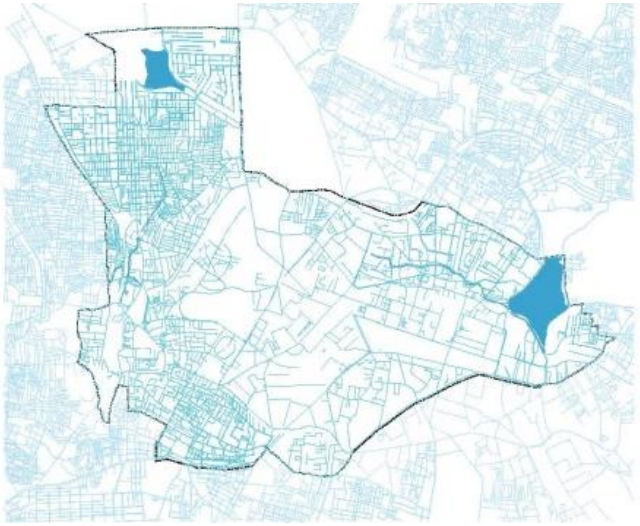
PARKS, OPEN SPACES & LAKES

13% Open space



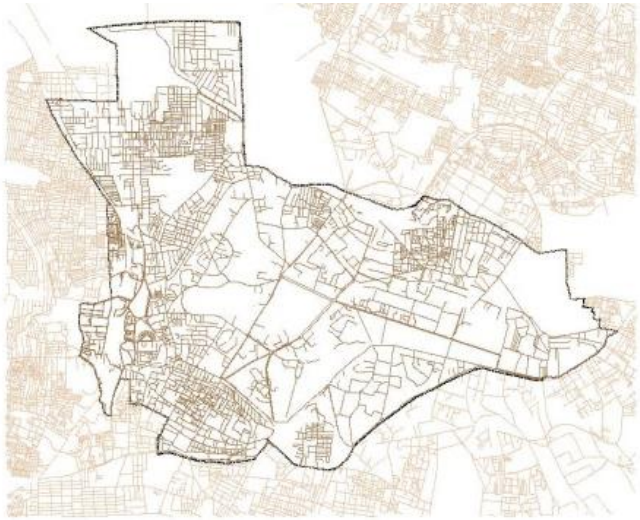
TRANSPORTATION NETWORK

97 km of major Roads



STORM WATER DRAIN NETWORK

556 km within ABD



SEWERAGE NETWORK

417 km within ABD

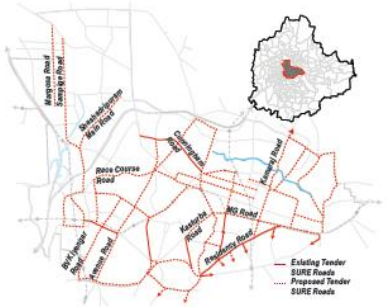
AREA BASED DEVELOPMENT

1. Revitalization Of Historic Heart Of City

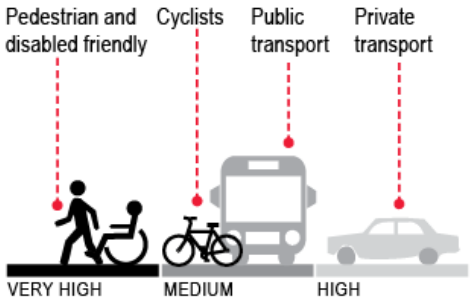
Redevelopment under Tender S.U.R.E.

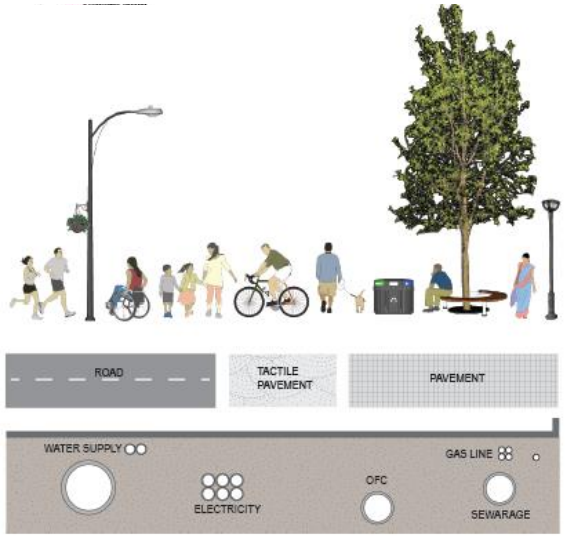
Proposed for 66% of Major roads in ABD

"…into a walking, cycling, tree lined, economically vibrant, environmentally sensible and inclusive center"



- **Infrastructure revitalization** (increase in capacity and coverage)- Surface drainage, sewage network, water supply, power and telecommunication
- The city core needs **highest degree of walkability** owing to highest footfall count





Tender S.U.R.E.	Promoting NMT	Improved public amenities	Energy efficiency/ conservation	Smart S.U.R.E.
Uniform travel lanes, underground services, smart parking, landscaping, street features	Pedestrian walkway and cycling network, app enabled public bicycle share points	Smart bus shelters with digital information boards, smart dustbins, environmental sensors, e-toilets, water ATMs, smart telecom towers	Vehicle charging points, Solar lighting/ energy efficient lighting, electric buses, e rickshaw stands with charging points	Based on PPP annuity model. Cost recovery on O&M from telecom companies

EXISTING PRECEDENT: 9 Roads – 11.5 km executed; 150 electric buses (BMTC proposal)

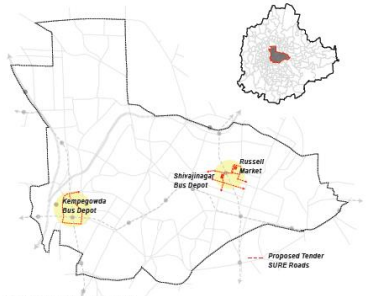
AREA BASED DEVELOPMENT

2. Integrated Mobility Towards Creating Vibrant Destination

Shivajinagar Bus Depot + Russell Market, Kempegowda Bus Stand

Integrated Development Of Transport Hub, Market, Civic Park & Connecting Road

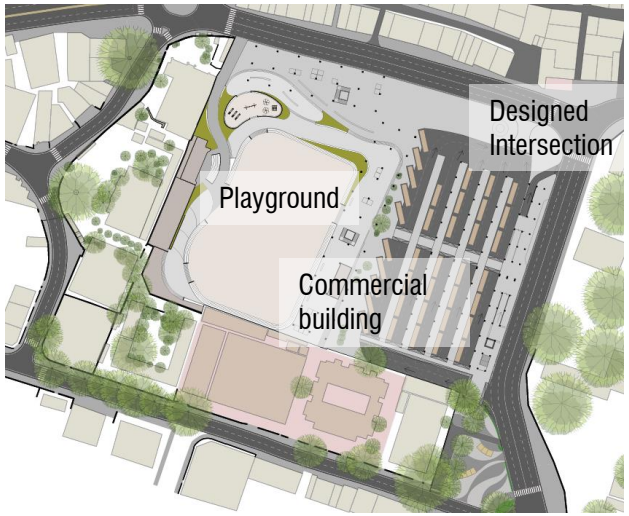
"Integrating transport hubs, vibrant market, school and community civic space towards a livelihood destination"



KEY MAP - MULTI MODAL HUBS

- **Building on Tender SURE and Proposed Metro Station**
- **Connecting Shivajinagar bus depot to Russell Market** through overhead pedestrian bridge
- **Reuse of underutilized public space and infrastructure** towards economic viability

Shivajinagar Context map



Shivajinagar Bus Depot Proposed Plan

Shivajinagar Bus Depot - Proposed

Redevelopment of Shivajinagar Bus Depot & Civic Park	Increased space allocation for economic activities	Integrated development of Russell Market	Improved parking in CBD	Retrofitting of Kempegowda Bus Stand (Majestic)
Bus routing in and around bus depot. Integrate metro network to bus transportation to increase regional connectivity	smart cubicles/ start-up hubs/ incubators within CBD, possibly on under utilized public holdings	Retrofitting of historic market building	App enabled smart car parking with vehicle charging points	Digital information boards, way finding features, CCTV surveillence

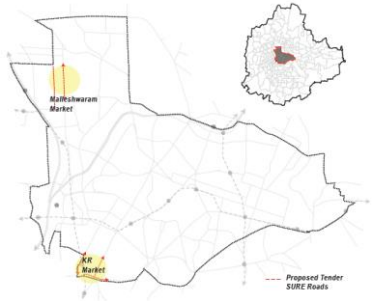
AREA BASED DEVELOPMENT

3. Upgradation & Redevelopment Of Historic Economic Centres

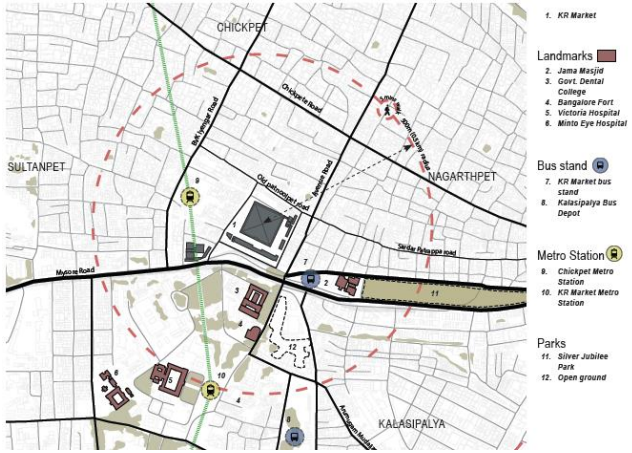
K R Market, Malleshwaram Market

Footfall of 30,000/ day, 1249 vendors

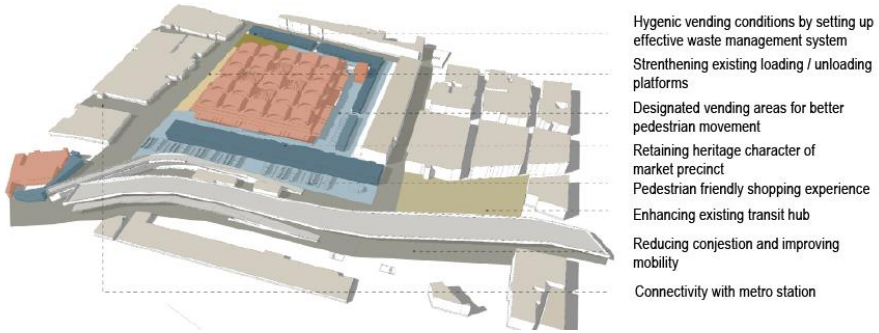
"Towards resilient markets: Connectivity with Metro Station, Reuse of existing unused municipal built assets and Solving the garbage crisis"



- **Retrofitting of historic market center** through improved public transport connectivity and efficient circulation management
- **Efficient vending and increased retail footprint** through efficient space management
- **Integrating proposed metro station to market** through over head pedestrian bridge



KR Market Context map



Proposals for market redesign

KR Market - Proposed entrance

Redesign of KR Market	Organized vending	Solid waste management	Energy efficiency/ conservation	Redevelopment of Malleshwaram Market
Heritage conservation, organizing vehicular and pedestrian movement around the market, provision of loading unloading bays and time managed goods movement	Creating designated vending zones with plazas – smart vendor kiosks	Sensor based smart dustbins, recycling unit, composting unit	Roof top solar panels, solar lighting/ energy efficient lighting	CCTV surveillance, e-toilets, water ATMs, Public bicycle share points

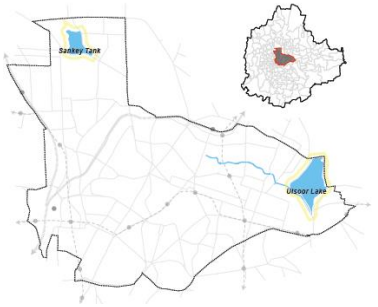
AREA BASED DEVELOPMENT

4. Innovation Of Downstream Clean Up Of Drainage System

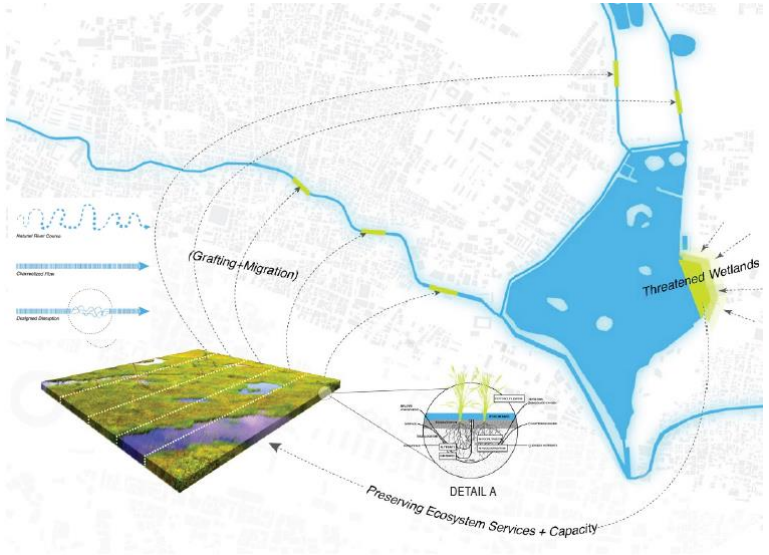
Ulsoor Lake, Sankey Lake

200 acres of lake precincts & 2.5 km of nallah clean up

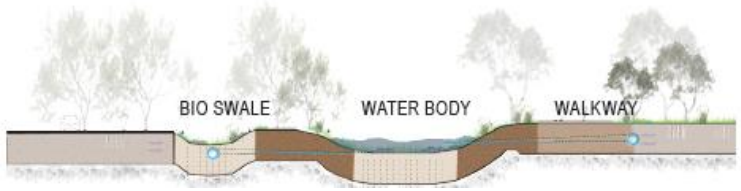
"Developing a water body as a replicable model towards reclaiming the dying historic drainage system of the city"



- **Green initiatives** for water body clean up
- **Increased ground water** recharge
- Creating a **vibrant and accessible public space** within the ABD



Initiatives for Ulsoor lake and nallah clean up



Bio swales along lake and nallah

DEWATs (Decentralised waste water treatment) for reducing contamination of water body

Proposed pedestrian and bicycle network around Ulsoor lake

Ulsoor lake and nallah clean up	Creating public space	Improved public amenities	Sankey tank
Introduction of DEWAT system, Green Bridges for natural filtration and Bio-Swales in lake and nallah for reducing water contamination	Lake front development as an active public space, Pedestrian and cycle network around the lake	Provision of e-toilets, Smart bins, Public space features/ furniture, public bicycle share points, energy efficient lighting, Solar trees and CCTV installation for improved security	Pedestrian and cycle network around the lake, smart features, children's play equipment, public space furniture

AREA BASED DEVELOPMENT

5. Protection & Redevelopment Of The Centrally Located Parkland

Cubbon Park

25000 foot fall / day (weekends)

"to become a local community asset"

- **Revitalizing the central city green** through improved pedestrian connectivity and place making
- **Integrate green network** to heritage and cultural landmarks around Cubbon park

Cubbon Park

6. Increasing Affordable Housing Stock Through Slum Redevelopment

Swathanthra Palya Slum (Gandhinagar)

Housing + road + Utility Services

"improving quality of life"

- Upgrading of **housing units**
- **physical infrastructure** - roads, water supply, sanitation, street lighting
- **social infrastructure -** Community centre

Proposed slum upgradation

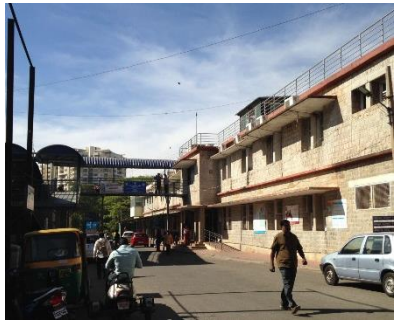
7. Retrofitting Of A Healtch Care Facility

K.C. General Hospital

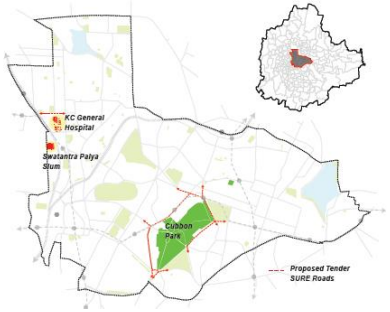
Building redesign + abutting road under Tender SURE

"improving a health care centre"

- **Building redesign** of health care centre
- **Heritage conservation** of building



Existing KC General Hospital



Cubbon Park	Safety and security	Rain water harvesting	Affordable housing	K.C. General hospital
Digital information board (activities, events, eateries and places of interest), tourist information kiosk, seating	CCTV Installation and improved energy efficient lighting	Collection and re use towards ground water recharge	Housing units, physical infrastructure, social infrastructure, roof top solar panel on community centre, energy efficient solar lighting	CCTV surveillance, Roof top solar panels, public bicycle share points, Tender SURE road for abutting

PAN CITY INTERVENTIONS

Project: Citizen Centric E-Governance and Services

PAN CITY PROPOSALS

Project 1: Citizen Centric E-governance And Services

1. Grievance Management — Governance

- Central Command Centre for accepting citizen queries across agencies
- Smart Kiosk for people without digital reach

2. Municipal Finance Reforms

- Municipal Audits
- City dashboard as a single MIS
- Surveying wards and streets to accurately report condition of infrastructure

3. Participatory Budgeting

- Campaigns and drive to involve residents and publish the collated, analysed results
- Citizen app that consolidates neighbourhood and ward level inputs from the citizens

4. Improvement of Neighborhood Safety

- Community Policing
- CCTV monitoring

5. Improvement in Property Tax Collections

- 3D building mapping for city using drone technology for property assessment, public assets, regulation and revenue augmentation

6. B-TRIPS (Bengaluru Transport related information and planning system)

- Single smart card for all public transport
- PIS system

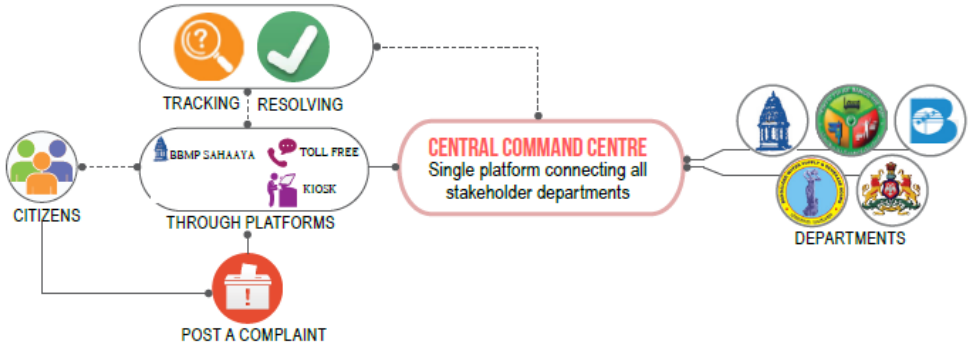
7. Public Project Management — Public Services

- Online project information system displaying projects information

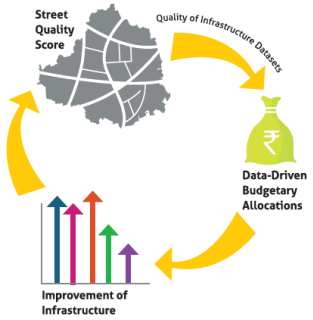
8. Data for Citizens

- Open data portal is based on the ideology of providing spatial data to citizens

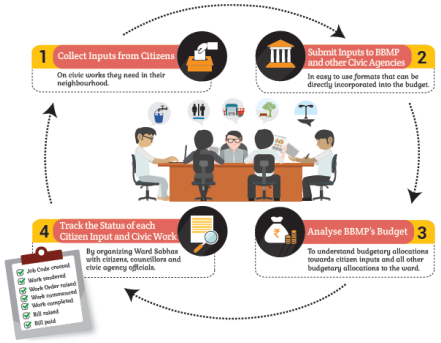
1. Grievance Management: Grievance redressal steps



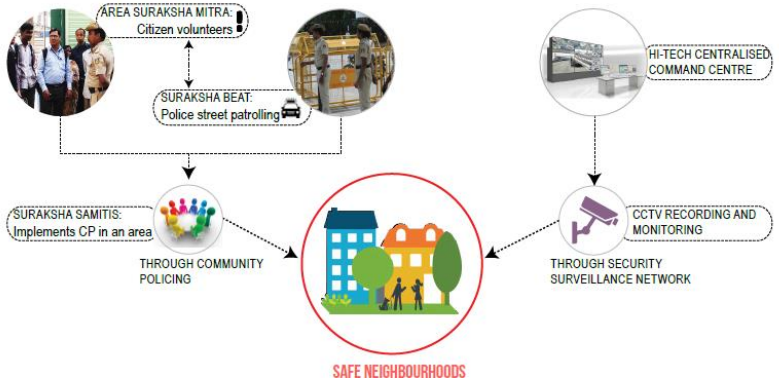
2. Municipal Finance Reforms: Street Quality Report Benefits



3. Participatory Budget



4. Key Initiatives for Safe Neighbourhoods



Thank You